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https://www.tensorflow.org/api_docs/python/tf/Operation

Represents a graph node that performs computation on tensors.

Function Signatures

```
tf.Operation( *args, **kwargs )
```

Code Examples

```
tf.Operation(  
    *args, **kwargs  
)
```

```
tf.Operation(  
    *args, **kwargs  
)
```

```
colocation_groups() -> list[bytes]
```

```
colocation_groups() -> list[bytes]
```

```
experimental_set_type(  
    type_proto  
) -> None
```

```
experimental_set_type(  
    type_proto  
) -> None
```

```
@classmethod  
from_node_def(  
    node_def,  
    g,  
    inputs=None,  
    output_types=None,  
    control_inputs=None,  
    input_types=None,  
    original_op=None,  
    op_def=None  
) -> OperationType
```

```
@classmethod
```

Documentation

Represents a graph node that performs computation on tensors.

View aliases

Compat aliases for migration

See

Migration guide for
more details.

`tf.compat.v1.Operation`

An Operation is a node in a `tf.Graph` that takes zero or more Tensor objects as input, and produces zero or more Tensor objects as output. Objects of type Operation are created by calling a Python op constructor (such as `tf.matmul`) within a `tf.function` or under a `tf.Graph.as_default` context manager.

For example, within a `tf.function`, `c = tf.matmul(a, b)` creates an Operation of type "MatMul" that takes tensors `a` and `b` as input, and produces `c` as output.

If a `tf.compat.v1.Session` is used, an Operation of a `tf.Graph` can be executed by passing it to `tf.Session.run`. `op.run()` is a shortcut for calling `tf.compat.v1.get_default_session().run(op)`.

Attributes

Before this op is executed, TensorFlow will ensure that the operations in `self.control_inputs` have finished executing. This mechanism can be used to run ops sequentially for performance reasons, or to ensure that the side effects of an op are observed in the correct order.

Methods

`## colocation_groups`

[View source](#)

Returns the list of colocation groups of the op.

experimental_set_type

[View source](#)

Sets the corresponding node's `experimental_type` field.

See the description of `NodeDef.experimental_type` for more info.

from_node_def

[View source](#)

Creates an Operation.

get_attr

[View source](#)

Returns the value of the attr of this op with the given name.

[View source](#)

Runs this operation in a Session.

Calling this method will execute all preceding operations that produce the inputs needed for this operation.

values

[View source](#)